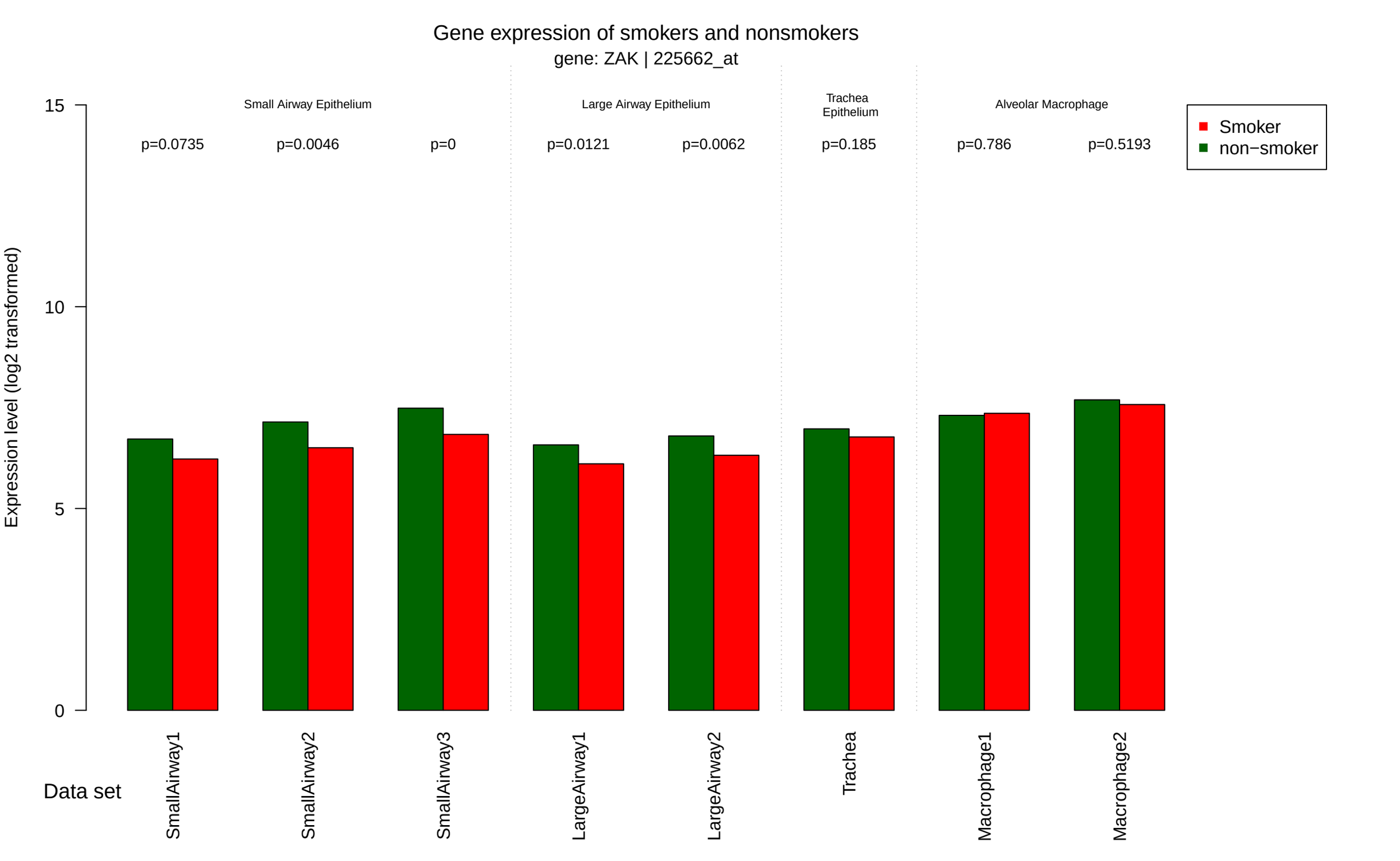
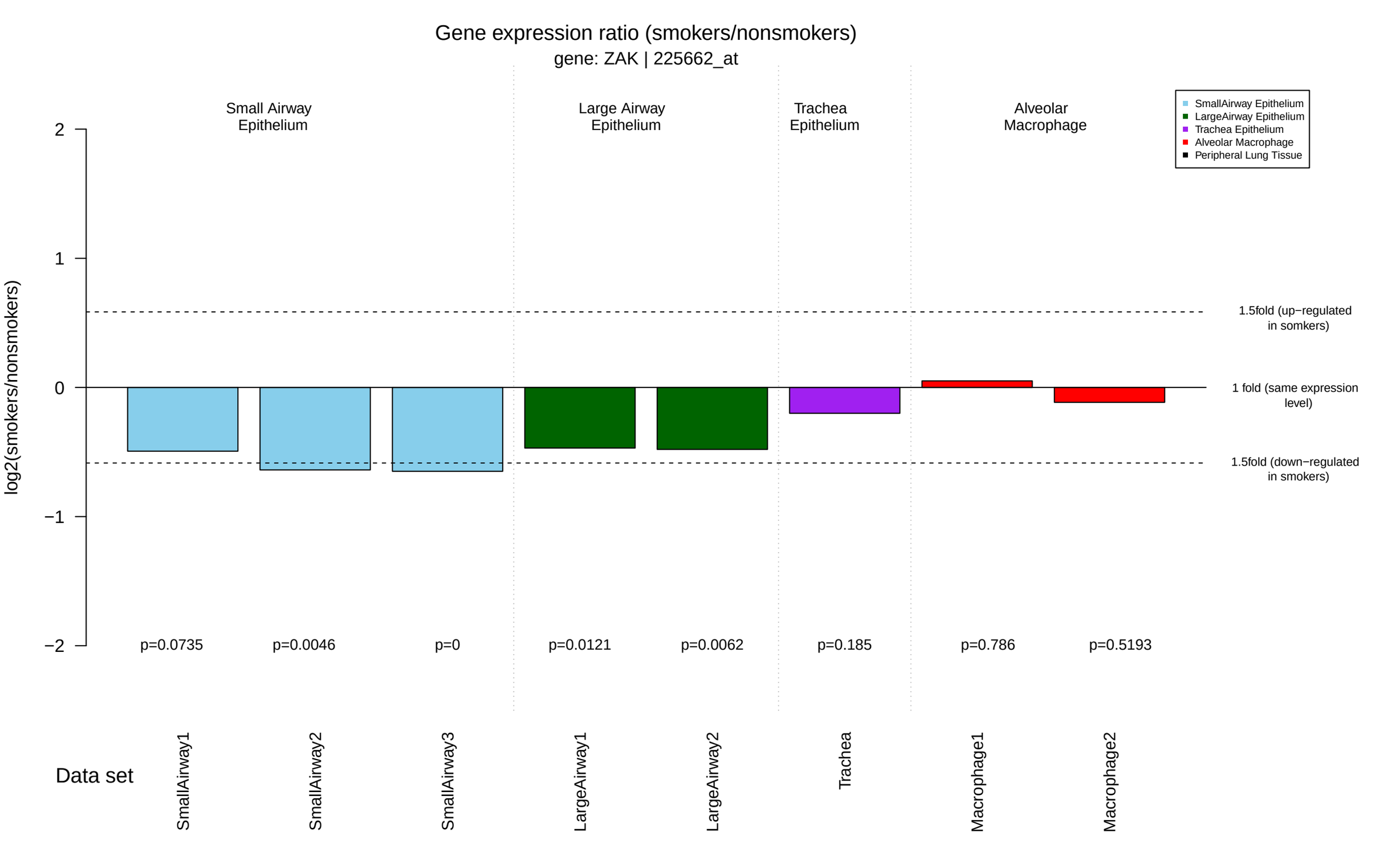
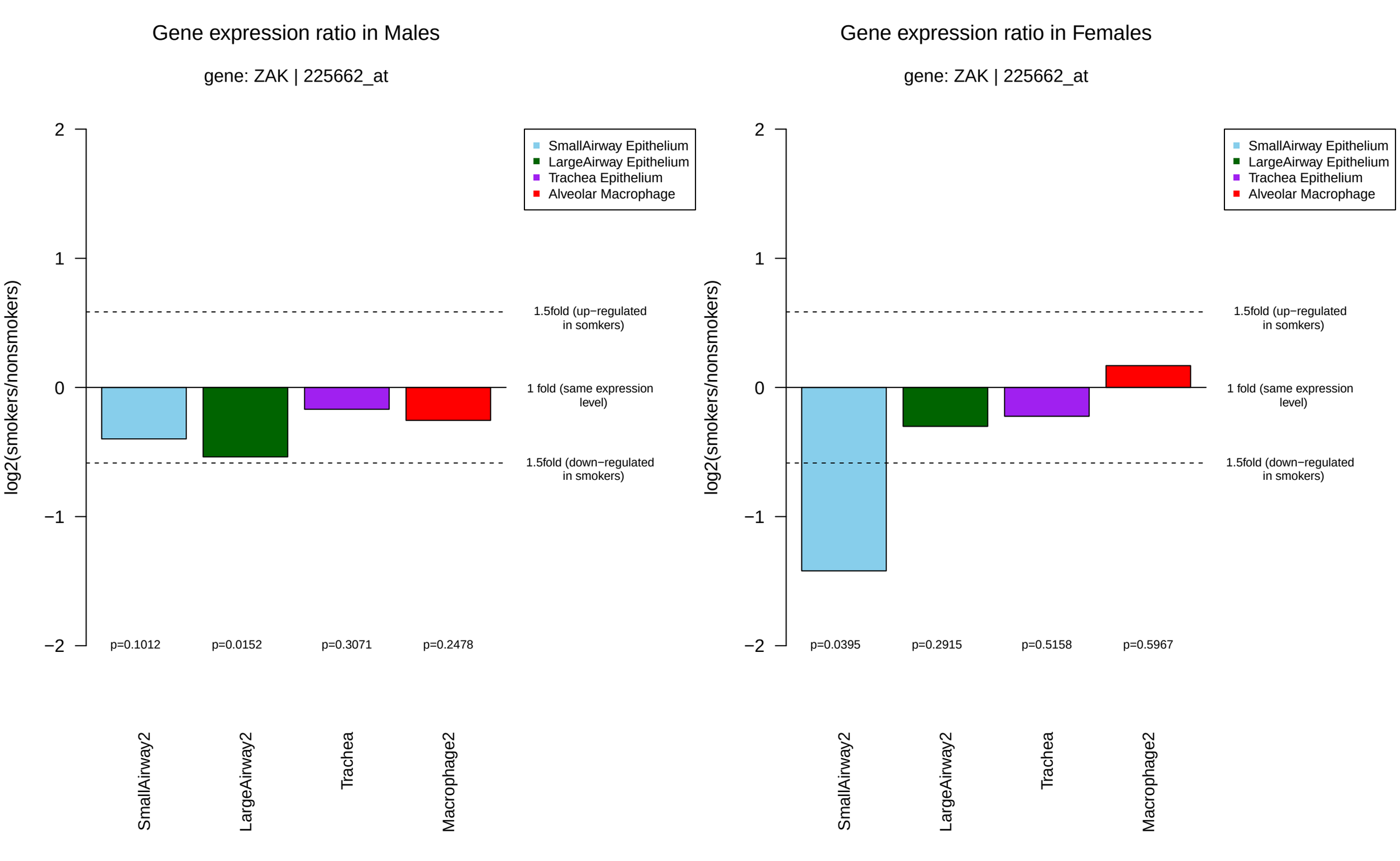
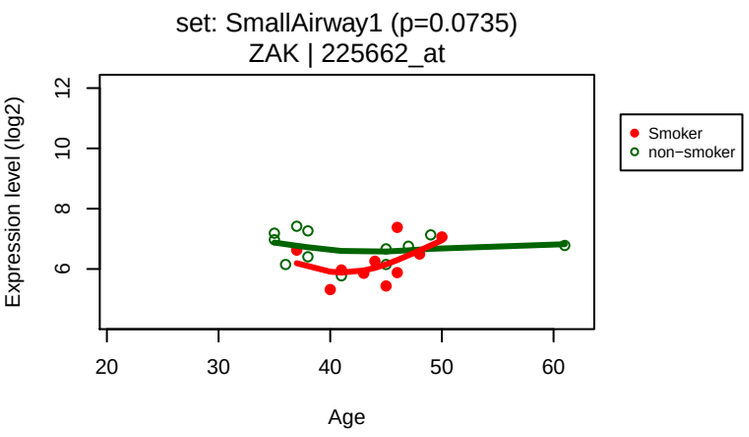
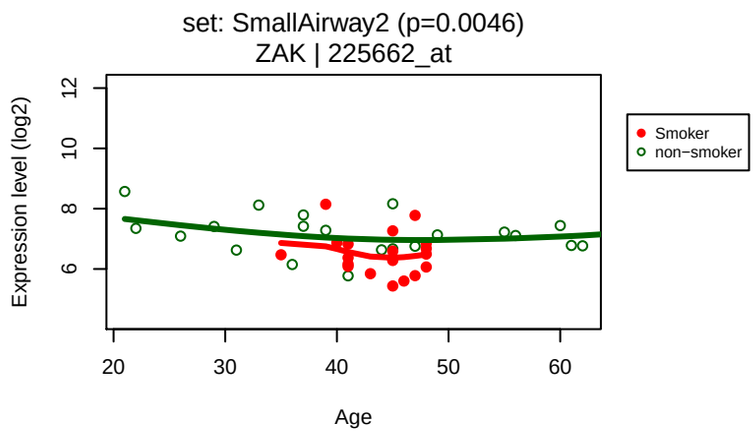
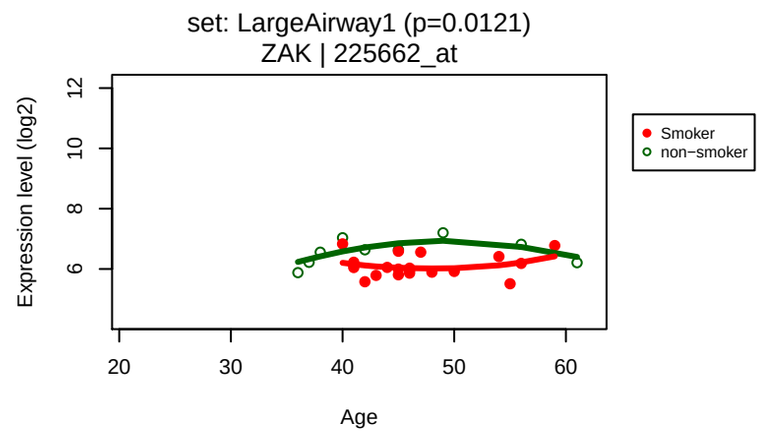

Gene expression of smokers and nonsmokers
gene: ZAK | 225662_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0735
p=0.0046
p=0
p=0.0121
p=0.0062
p=0.185
p=0.786
p=0.5193
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

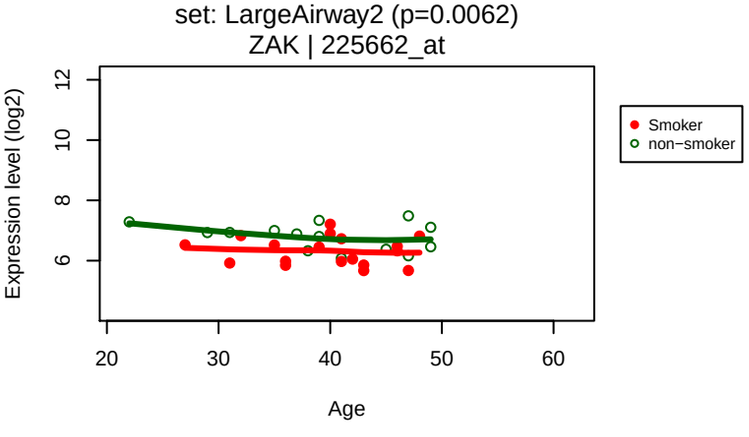

Gene expression ratio (smokers/nonsmokers)
gene: ZAK | 225662_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0735
p=0.0046
p=0
p=0.0121
p=0.0062
p=0.185
p=0.786
p=0.5193
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

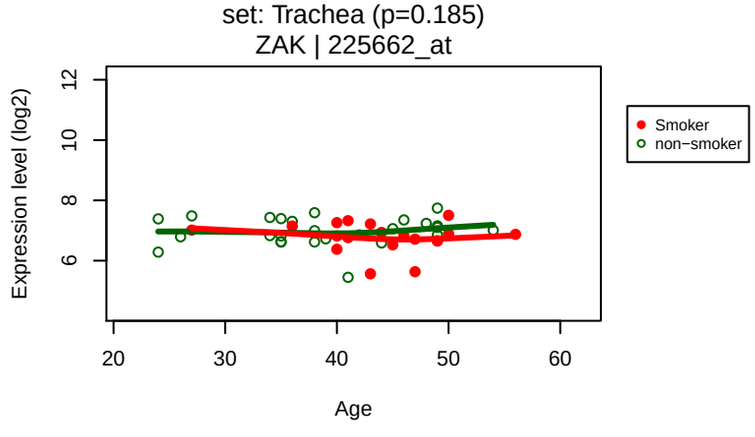

Gene expression ratio in Males
gene: ZAK | 225662_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1012
p=0.0152
p=0.3071
p=0.2478
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ZAK | 225662_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0395
p=0.2915
p=0.5158
p=0.5967
SmallAirway2
LargeAirway2
Trachea
Macrophage2

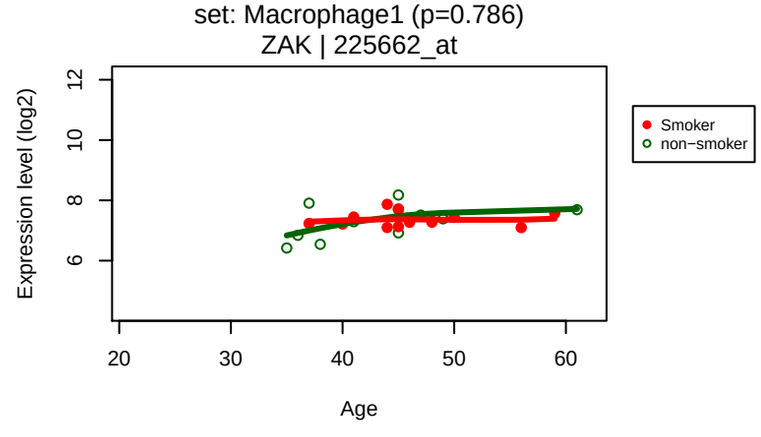

set: SmallAirway1 (p=0.0735)
ZAK | 225662_at
Expression level (log2)
Age
Smoker
non−smoker

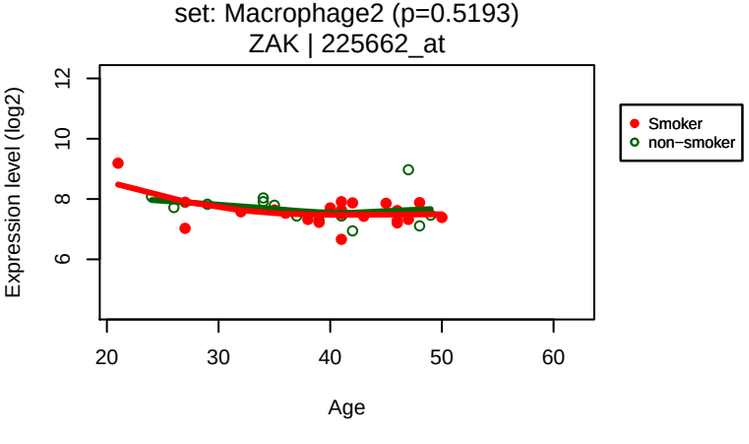

set: SmallAirway2 (p=0.0046)
ZAK | 225662_at
Expression level (log2)
Age
Smoker
non−smoker

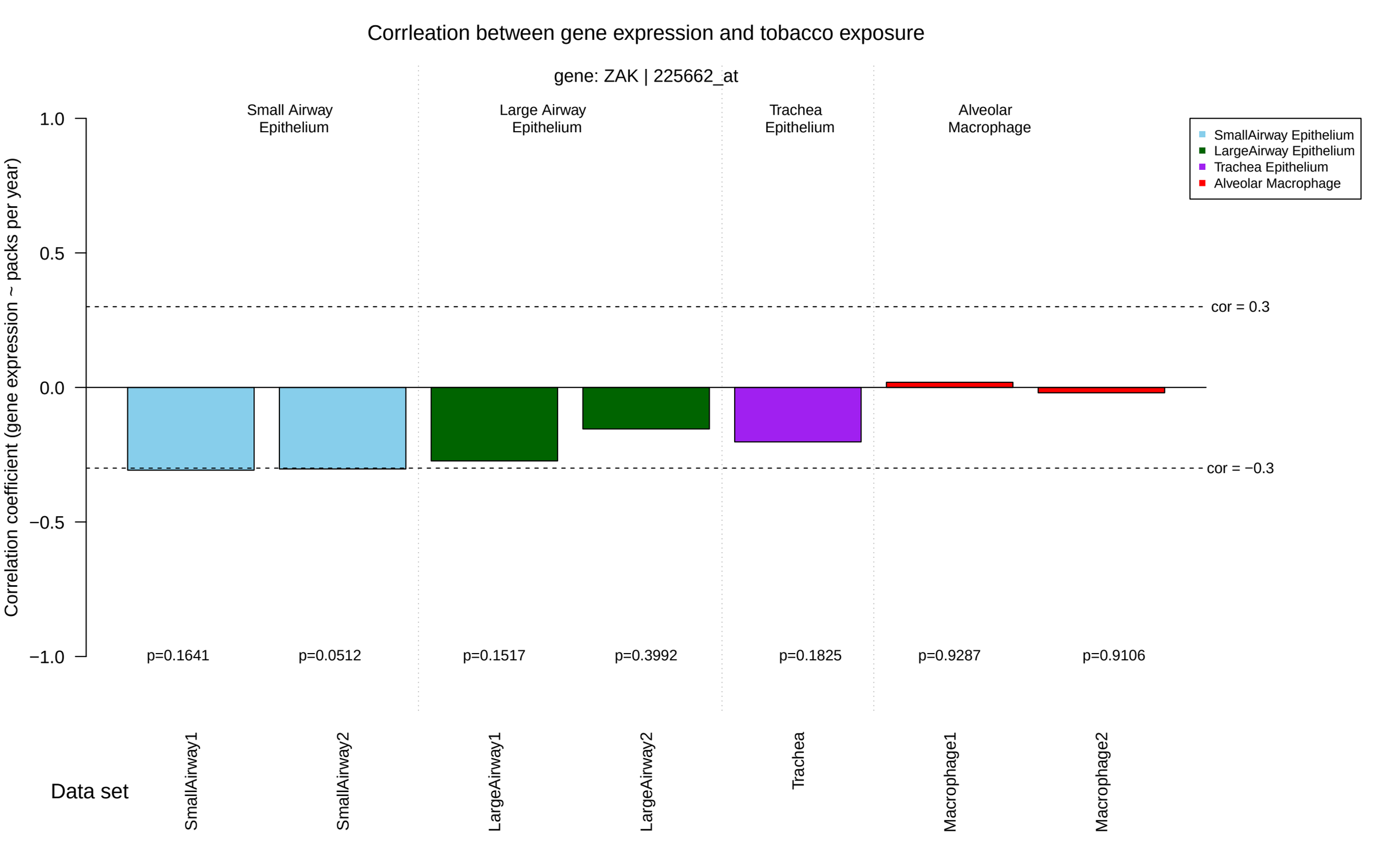

set: LargeAirway1 (p=0.0121)
ZAK | 225662_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0062)
ZAK | 225662_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.185)
ZAK | 225662_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.786)
ZAK | 225662_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5193)
ZAK | 225662_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ZAK | 225662_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1641
p=0.0512
p=0.1517
p=0.3992
p=0.1825
p=0.9287
p=0.9106
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2